

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Logic Reasoning s Mapping Exercise

COURSE NAME: ARTIFICIAL INTELLIGENCE **COURSE CODE:** MLA01
AND EXPERT SYSTEMS

COURSE OUTCOME COVERED

CO3: Implement propositional and first-order logic using resolution, unification, and inference techniques for knowledge representation and reasoning.(BTL 3)
Assessment Tool Weightage: 15%

Total Marks: 100

Time: c0 Mins

No. of Questions: 5

Annexure A

Logic Reasoning s Mapping Exercise

Code of Conduct:

I N.Sri Balaji-192525275 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *N.Sri Balaji*

Criterion	Weightage	Marks Obtained
1. Problem Analysis and Knowledge Identification	15%	
2. Logical Representation and Mapping	20%	
3. Predicates, Variables, Constants and Quantifiers	15%	
4. Logical Connectives and Formula Construction	10%	
5. Inference and Logical Reasoning	15%	
6. Unification and Substitution	10%	
7. Resolution and Proof Construction	10%	
8. Mapping Diagram / Clarity and Presentation	5%	
Total	100%	

Smart Healthcare Diagnosis

Given Scenario

Ravi has fever, cough, and body pain. Patients with fever and cough may have flu. Patients with flu and body pain may have viral infection. Patients with viral infection require medical attention.

(i) Facts, Entities, Predicates, Variables and Constants

Entity

The main entity in the scenario is:

- Patient: Ravi

Constant

- Ravi

Facts

- Ravi has fever.
- Ravi has cough.
- Ravi has body pain.

Predicates

Let:

- $\text{Fever}(x) = x$ has fever.
- $\text{Cough}(x) = x$ has cough.
- $\text{BodyPain}(x) = x$ has body pain.
- $\text{Flu}(x) = x$ has flu.
- $\text{ViralInfection}(x) = x$ has viral infection.
- $\text{MedicalAttention}(x) = x$ requires medical attention.

Variables

- x represents any patient.

Rules

1. If a patient has fever and cough, then the patient may have flu.
2. If a patient has flu and body pain, then the patient may have viral infection.
3. If a patient has viral infection, then the patient requires medical attention.

(ii) Propositional Logic Representation

Let:

- F = Ravi has fever.
- C = Ravi has cough.
- B = Ravi has body pain.
- L = Ravi has flu.
- V = Ravi has viral infection.
- M = Ravi requires medical attention.

The given facts are:

$F = \text{True}$

$C = \text{True}$

$B = \text{True}$

The rules can be represented as:

1. Fever and cough imply flu:

$F \wedge C \rightarrow L$

2. Flu and body pain imply viral infection:

$L \wedge B \rightarrow V$

3. Viral infection implies medical attention:

$$V \rightarrow M$$

Therefore, the complete propositional knowledge base is:

$$F \wedge C$$

$$F \wedge C \rightarrow L$$

$$L \wedge B \rightarrow V$$

$$V \rightarrow M$$

(iii) First-Order Logic Representation

The knowledge can be represented using predicates and universal quantifiers.

Facts

Fever(Ravi)

Cough(Ravi)

BodyPain(Ravi)

Rules

1. For every patient, if the patient has fever and cough, then the patient has flu:

$$\forall x [(Fever(x) \wedge Cough(x)) \rightarrow Flu(x)]$$

2. For every patient, if the patient has flu and body pain, then the patient has viral infection:

$$\forall x [(Flu(x) \wedge BodyPain(x)) \rightarrow ViralInfection(x)]$$

3. For every patient, if the patient has viral infection, then the patient requires medical attention:

$$\forall x [ViralInfection(x) \rightarrow MedicalAttention(x)]$$

Here, $\forall x$ means “for every patient x.”

(iv) Unification and Substitution

Consider the predicates:

Fever(x)

and

Fever(Ravi)

To make both predicates identical, substitute:

$$x = \text{Ravi}$$

Therefore, the Most General Unifier (MGU) is:

$$\{x/\text{Ravi}\}$$

After substitution:

$$Fever(x) \rightarrow Fever(\text{Ravi})$$

Similarly, consider:

Flu(x)

and

Flu(Ravi)

The substitution is:

$$\{x/\text{Ravi}\}$$

Therefore:

$$Flu(x)\{x/\text{Ravi}\} = Flu(\text{Ravi})$$

Thus, the predicates are successfully unified.

(v) Resolution to Prove Ravi Has Viral Infection

We need to prove:

ViralInfection(Ravi)

The relevant facts are:

Fever(Ravi)

Cough(Ravi)

BodyPain(Ravi)

The rules are:

$\text{Fever}(x) \wedge \text{Cough}(x) \rightarrow \text{Flu}(x)$

$\text{Flu}(x) \wedge \text{BodyPain}(x) \rightarrow \text{ViralInfection}(x)$

Step 1: Derive Flu(Ravi)

From:

Fever(Ravi)

and

Cough(Ravi)

Using the rule:

$\text{Fever}(x) \wedge \text{Cough}(x) \rightarrow \text{Flu}(x)$

Substitute:

$x = \text{Ravi}$

Therefore:

Flu(Ravi)

Step 2: Derive ViralInfection(Ravi)

We already know:

Flu(Ravi)

and:

BodyPain(Ravi)

Using the rule:

$\text{Flu}(x) \wedge \text{BodyPain}(x) \rightarrow \text{ViralInfection}(x)$

Substitute:

$x = \text{Ravi}$

Therefore:

ViralInfection(Ravi)

Hence, Ravi has a viral infection according to the given knowledge base.

(vi) Complete Inference Sequence and Final Conclusion

The complete reasoning sequence is:

Fact 1: Ravi has fever.

Fact 2: Ravi has cough.

Fact 3: Ravi has body pain.

Inference 1:

$\text{Fever}(\text{Ravi}) \wedge \text{Cough}(\text{Ravi})$

$\rightarrow \text{Flu}(\text{Ravi})$

Therefore, Ravi has flu.

Inference 2:

$\text{Flu}(\text{Ravi}) \wedge \text{BodyPain}(\text{Ravi})$

$\rightarrow \text{ViralInfection}(\text{Ravi})$

Therefore, Ravi has a viral infection.

Inference 3:

$\text{ViralInfection}(\text{Ravi})$

$\rightarrow \text{MedicalAttention}(\text{Ravi})$

Therefore, Ravi requires medical attention.

Final Conclusion

Ravi has a viral infection and requires medical attention.

The inference is obtained logically from the given facts and rules using unification, substitution, and logical inference.

